

EE5239 Introduction to Nonlinear Optimization

Lecture 7b: Lagrangian Multipliers (Equality Constraints)

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Equality Constrained Problem

Let us first consider the following equality constrained problem

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0, \quad i = 1, \dots, m. \end{array}$$

where $f : R^n \mapsto R$, $h_i : R^n \mapsto R, i = 1, \dots, m$, are continuously differentiable function.

n variables, m constraints

Lagrangian Function

- The Lagrangian function can be formed using the Lagrangian multipliers $\nu_i \in \mathbb{R}$

$$L(x, \nu) = f(x) + \underbrace{\sum_{i=1}^m \nu_i h_i(x)}_{\text{equality constraints}}$$

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- **Question:** Can we use Lagrangian function to characterize the optimality conditions?

Equality Constrained Problem

Lagrange Multiplier Theorem (necessary condition)

- Let x^* be a local min and a regular point [the vectors of $\nabla h_i(x^*)$'s are all linearly independent]. Then there exist unique scalars $\lambda_1^*, \dots, \lambda_m^*$ such that

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*) = 0.$$

- If in addition f and h are twice continuously differentiable,

$$y' \left(\nabla^2 f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla^2 h_i(x^*) \right) y \geq 0, \quad \forall y \text{ s.t. } \nabla h_i(x^*)' y = 0, \quad \forall i$$

Characterizes the first-order, and second-order necessary conditions for local min.

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$$\nabla f(x^*) = - \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*).$$

- Interpretation I.** At a local optimal solution, the gradient of the function can be expressed as a linear combination of the gradients of the constraints

Question: Is this always possible to find these $\{\lambda_i^*\}$?

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- Let x^* be a local min and a **regular point**[the vectors of $\nabla h_i(x^*)$'s are all linearly independent]. Then there exist unique scalars $\lambda_1^*, \dots, \lambda_m^*$ such that

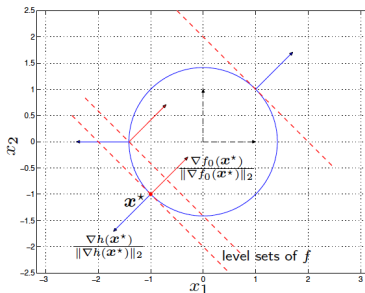
$$\nabla f(x^*) = - \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*).$$

- Interpretation I.** At a local optimal solution, the gradient of the function can be expressed to a **linear combination** of the gradients of the constraints
- Interpretation II.** The cost gradient $\nabla f(x^*)$ is orthogonal to the subspace of **first order feasible variations**

$$V(x^*) = \{\Delta x \mid \nabla h_i(x^*)' \Delta x = 0, \forall i\} \quad (0.1)$$

Question: Is this always possible to find these $\{\lambda_i^*\}$?

Example 1



Consider the problem

$$\underset{\mathbf{x} \in \mathbb{R}^2}{\text{minimize}} \quad f_0(\mathbf{x}) = x_1 + x_2$$

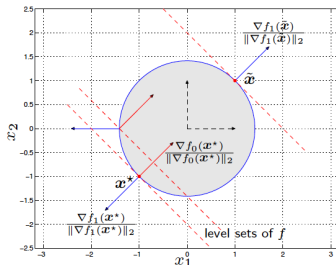
$$\text{subject to} \quad h(\mathbf{x}) = x_1^2 + x_2^2 - 2 = 0.$$

This is a problem with a linear objective function $f(\mathbf{x})$ and one nonlinear equality constraint $h(\mathbf{x}) = 0$. At the solution $\mathbf{x}^*$, the gradient of the constraint $\nabla h(\mathbf{x}^*)$ is orthogonal to the level set of the function at $\mathbf{x}^*$, and hence $\nabla h(\mathbf{x}^*)$ and $\nabla f_0(\mathbf{x}^*)$ are parallel *i.e.*, there is a scalar ν^* such that

$$\nabla f_0(\mathbf{x}^*) + \nu^* \nabla h(\mathbf{x}^*) = \mathbf{0}.$$

Clearly, in this example $\mathbf{x}^*$ is regular (because $\nabla h(\mathbf{x}^*) \neq \mathbf{0}$).

Example 2



Consider the problem

$$\text{minimize}_{\mathbf{x} \in \mathbb{R}^2} f_0(\mathbf{x}) = x_1 + x_2$$

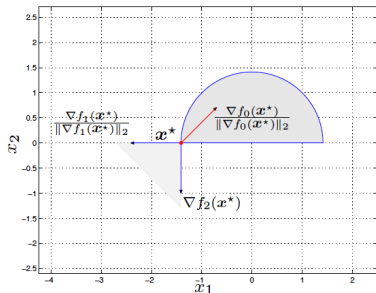
$$\text{subject to } f_1(\mathbf{x}) = x_1^2 + x_2^2 - 2 \leq 0.$$

This is a problem with a linear objective function $f(\mathbf{x})$ and one nonlinear inequality constraint $f_1(\mathbf{x}) \leq 0$. At the solution $\mathbf{x}^*$, the gradient of the constraint $\nabla f_1(\mathbf{x}^*)$ is orthogonal to the level set of the function at $\mathbf{x}^*$, and the following equality holds

$$\nabla f_0(\mathbf{x}^*) + \lambda^* \nabla f_1(\mathbf{x}^*) = \mathbf{0},$$

for $\lambda^* = \frac{1}{2} \geq 0$. Note that at the point $\tilde{\mathbf{x}} = (1, 1)$, $\nabla f_0(\tilde{\mathbf{x}}) + \lambda \nabla f_1(\tilde{\mathbf{x}}) = \mathbf{0}$ holds as well, however $\lambda = -\frac{1}{2} \leq 0$.

Example 3



Consider the problem

$$\begin{aligned} & \underset{x \in \mathbb{R}^2}{\text{minimize}} && f_0(x) = x_1 + x_2 \\ & \text{subject to} && f_1(x) = x_1^2 + x_2^2 - 2 \leq 0, \\ & && f_2(x) = -x_2 \leq 0. \end{aligned}$$

At the solution $x^* = (-\sqrt{2}, 0)$, $-\nabla f_0(x^*)$ belongs to the normal cone to the feasible set at point x^* , hence, there is $\lambda^* \geq 0$ that satisfies

$$\nabla f_0(x^*) + \lambda_1^* \nabla f_1(x^*) + \lambda_2^* \nabla f_2(x^*) = 0.$$

Example 4

Consider the problem

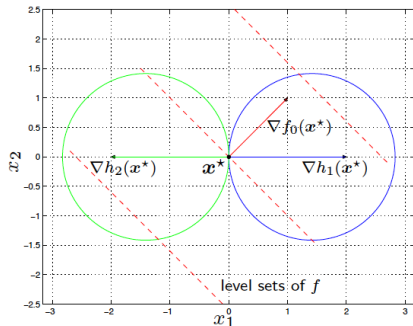
$$\underset{\mathbf{x} \in \mathbb{R}^2}{\text{minimize}} \quad f_0(\mathbf{x}) = x_1 + x_2$$

$$\text{subject to} \quad h_1(\mathbf{x}) = (x_1 + 1)^2 + x_2^2 - 2 = 0.$$

$$h_2(\mathbf{x}) = (x_1 - 1)^2 + x_2^2 - 2 = 0.$$

This is a problem with a linear objective function $f_0(\mathbf{x})$ and two nonlinear equality constraints $h_1(\mathbf{x}) = 0$, $h_2(\mathbf{x}) = 0$. At the solution $\mathbf{x}^*$ (which is in fact the only feasible point), no linear combination of the gradients of the two constraints is equal to $\nabla f_0(\mathbf{x}^*)$, i.e., there is no $\boldsymbol{\nu}^*$ such that

$$\nabla f_0(\mathbf{x}^*) + \nu_1^* \nabla h_1(\mathbf{x}^*) + \nu_2^* \nabla h_2(\mathbf{x}^*) = \mathbf{0}.$$



Summary

- From these examples, we know that the first-order necessary condition we just mentioned (by using Lagrangian multipliers to characterize) may or may not hold true
- That is the reason why, in the previous statement, we have to make the following critical assumption [regularity condition]:
The vectors of $\nabla h_i(x^*)$'s are all linearly independent.
- Next we are going into proof steps of the first-order conditions.
- Assuming x^* is a local min, we will show **exists** λ^* such that first-order condition holds true

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$$f^k(x) = f(x) + k|h(x)|^2 + \frac{\alpha}{2} \|x - x^*\|^2, \quad k = 1, 2, \dots$$

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- Let x^k be a constrained global minimizer of f^k over the region

$$\{x \mid \|x - x^*\| \leq \delta\}.$$

We will show that x^k is an unconstrained local min of f^k for all large k .

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- Taking limit $k \rightarrow \infty$ of

$$f^k(x^k) = f(x^k) + k|h(x^k)|^2 + \frac{\alpha}{2} \|x^k - x^*\|^2 \leq f^k(x^*) = f(x^*)$$

along **any** convergent subsequence of $\{x^k\}$.

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- The above is true because x^* is a feasible solution to

$$\min f^k(x) \tag{0.2}$$

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- Further, let $x^k \rightarrow \bar{x}$ be any of its limit point;
 $\lim_{k \rightarrow \infty} f^k(x^k) \leq f(x^*)$ implies $k|h(x^k)|$ is bounded as $k \rightarrow \infty$,
so the following holds

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- Thus, we have $\bar{x} = x^*$ and $f(x^*) = f(\bar{x})$.

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- Since $\bar{x}$ is any limit point, we have $x^k \rightarrow x^*$;
- So $\|x^k - x^*\| < \delta$ for large k ; This implies for k large enough, x^k is an unconstrained min of f^k [that is, x^k will not touch the boundary of $\|x^k - x^*\| = \delta$], so it satisfies

$$\nabla f^k(x^k) = 0, \quad \nabla^2 f^k(x^k) \succeq 0.$$

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- Taking limit of the following optimality condition

$$0 = \nabla f(x^k) + 2k \nabla h(x^k) h(x^k) + \alpha(x^k - x^*) \quad (0.4)$$

Where $\nabla h(x^*)$ is a size $n \times m$ matrix of the gradient of the constraints (also known as the Jacobian matrix)

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Where $\nabla h(x^*)$ is a size $n \times m$ matrix of the gradient of the constraints (also known as the Jacobian matrix)

- $\nabla h(x^*)$ has rank m , $\nabla h(x^k)$ also has rank m for large k , so $\nabla h(x^k)' \nabla h(x^k)$: invertible.

Proof of the Theorem (Sec. 3.1.1./4.1.1)

- Recall from the previous slides

$$0 = \nabla f(x^k) + 2k \nabla h(x^k) h(x^k) + \alpha(x^k - x^*) \quad (0.5)$$

- Multiplying both sides with $\nabla h(x^k)'$, and taking the inverse, yields

$$k h(x^k) = -(\nabla h(x^k)' \nabla h(x^k))^{-1} \nabla h(x^k)' (\nabla f(x^k) + \alpha(x^k - x^*)).$$

- Taking limit as $k \rightarrow \infty$ and $x^k \rightarrow x^*$,

$$\{k h(x^k)\} \rightarrow (\nabla h(x^*)' \nabla h(x^*))^{-1} \nabla h(x^*)' \nabla f(x^*) \equiv \lambda.$$

Proof of the Theorem (Sec. 3.1.1./4.1.1)

Taking limit as $k \rightarrow \infty$ in Eq.(4), we obtain

$$\nabla f(x^*) + \nabla h(x^*)\lambda = 0.$$

This completes the proof.

Lagrangian Function

- Define the Lagrangian function

$$L(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i h_i(x).$$

Then, if x^* is a local minimum which is regular, the **Lagrange multiplier conditions** can be written as (why the second equality?)

$$\nabla_x L(x^*, \lambda^*) = 0, \quad \nabla_\lambda L(x^*, \lambda) = 0.$$

- System of $n + m$ equations with $n + m$ unknowns.
- 2nd order necessary condition:

$$y' \nabla_{xx}^2 L(x^*, \lambda^*) y \geq 0, \quad \forall y \text{ s.t. } \nabla h(x^*)' y = 0.$$

Lagrangian Function

- Example:

$$\begin{array}{ll}\text{minimize} & \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) \\ \text{subject to} & x_1 + x_2 + x_3 = 3.\end{array}$$

Necessary conditions

$$x_1^* + \lambda^* = 0, \quad x_2^* + \lambda^* = 0, \quad x_3^* + \lambda^* = 0, \quad x_1^* + x_2^* + x_3^* = 3.$$

Sufficiency Condition

- 2nd Order Sufficiency Conditions: Let $x^* \in R^n$ and $\lambda \in R^m$ satisfy

$$\nabla_x L(x^*, \lambda^*) = 0, \quad \nabla_\lambda L(x^*, \lambda^*) = 0 \quad (0.6)$$

$$y' \nabla_{xx}^2 L(x^*, \lambda^*) y > 0, \quad \forall y \neq 0 \text{ with } \nabla h(x^*)' y = 0. \quad (0.7)$$

Then x^* is a strict local minimum.

Sufficiency Condition

- Example:

$$\begin{array}{ll}\text{minimize} & -(x_1x_2 + x_2x_3 + x_1x_3) \\ \text{subject to} & x_1 + x_2 + x_3 = 3.\end{array}$$

We have that $x_1^* = x_2^* = x_3^* = 1$ and $\lambda^* = 2$ satisfy the 1st order conditions. Also

$$\nabla_{xx}^2 L(x^*, \lambda^*) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

We have for all $y \neq 0$ with $\nabla h(x^*)'y = 0$ or $y_1 + y_2 + y_3 = 0$,

$$\begin{aligned} y' \nabla_{xx}^2 L(x^*, \lambda^*) y &= -y_1(y_2 + y_3) - y_2(y_1 + y_3) - y_3(y_1 + y_2) \\ &= y_1^2 + y_2^2 + y_3^2 > 0 \end{aligned} \tag{0.8}$$

Hence, x^* is a strict local minimum.

A Useful Lemma [read]

- Let P and Q be two symmetric matrices. Assume that $Q \succ 0$ and $P \succ 0$ on the nullspace of Q , i.e., $x'Px > 0$ for all $x \neq 0$ with $x'Qx = 0$. Then there exists a scalar c such that

$$P + cQ : \text{positive definite, } \forall c > \bar{c}.$$

- Proof: Assume the contrary. Then for every k , there exists a vector x^k with $x^k = 1$ such that

$$(x^k)'Px^k + k(x^k)'Qx^k < 0.$$

Consider a subsequence $\{x^k\}_{k \in \mathcal{K}}$ converging to some x with $x = 1$. Taking the limit superimum,

$$x'Px + \limsup_{k \rightarrow \infty} \limsup_{k \in \mathcal{K}} (k(x^k)'Qx^k) \leq 0.$$

We have $(x^k)'Qx^k \geq 0$ (since $Q \succeq 0$), so

$$\{(x^k)'Qx^k\}_{k \in \mathcal{K}} \rightarrow 0.$$

Therefore, $x'Qx = 0$ and using the hypothesis, $x'Px > 0$, a contradiction.

Proof of Sufficiency Conditions [read]

Consider the *augmented Lagrangian function*

$$L_c(x, \lambda) = f(x) + \lambda'h(x) + \frac{c}{2} \|h(x)\|^2,$$

where c is a scalar. We have

$$\nabla_x L_c(x, \lambda) = \nabla_x L(x, \tilde{\lambda}), \tag{0.9}$$

$$\nabla_{xx}^2 L_c(x, \lambda) = \nabla_{xx}^2 L(x, \tilde{\lambda}) + c \nabla h(x) \nabla h(x)' \tag{0.10}$$

where $\tilde{\lambda} = \lambda + ch(x)$.

Proof of Sufficiency Conditions [read]

If (x^*, λ^*) satisfy the sufficiency conditions, we have using the lemma (why?),

$$\nabla_x L_c(x^*, \lambda^*) = 0, \quad \nabla_{xx}^2 L_c(x^*, \lambda^*) > 0,$$

for sufficiently large c . Hence for some $\gamma > 0, \epsilon > 0$, $(x^*$
unconstrained local min)

$$L_c(x, \lambda^*) \geq L_c(x^*, \lambda^*) + \frac{\gamma}{2} \|x - x^*\|^2,$$

if $\|x - x^*\| < \epsilon$. Since $L_c(x, \lambda^*) = f(x)$ when $h(x) = 0$,

$$f(x) \geq f(x^*) + \frac{\gamma}{2} \|x - x^*\|^2, \quad \text{if } h(x) = 0, \|x - x^*\| < \epsilon.$$

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- Since $b + \Delta b = a'(x^* + \Delta x) = a'x^* + a'\Delta x = b + a'\Delta x$, we have $a'\Delta x = \Delta b$. Using the condition $\nabla f(x^*) = -\lambda^* a$,
$$\Delta cost = f(x^* + \Delta x) - f(x^*) = \nabla f(x^*)' \Delta x + o(\|\Delta x\|) = -\lambda^* a' \Delta x + o(\|\Delta x\|)$$

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- Thus $\Delta cost = -\lambda^* \Delta b + o(\|\Delta x\|)$, so up to first order
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- For multiple constraints $a'_i x = b_i$, $i = 1, \dots, n$, we have

$$\Delta cost = -\sum_{i=1}^m \lambda_i^* \Delta b_i + o(\|\Delta x\|).$$